Math 241, F12 Name: Answer

Quiz 11 SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (5 points) Evaluate the iterated integral $\int_0^{\frac{\pi}{2}}\int_0^x\int_0^y \cos(x+y+z)\, dzdydx$.

$$\int_0^{\frac{\pi}{2}}\int_0^x\int_0^y \cos(x+y+z)\,dzdydx$$

$$= \int_0^{\frac{\pi}{2}}\int_0^x \Big[\sin(x+y+z)\Big]_{z=0}^{z=y} dydx = \int_0^{\frac{\pi}{2}}\int_0^x \sin(x+2y)-\sin(x+y)\,dydx$$

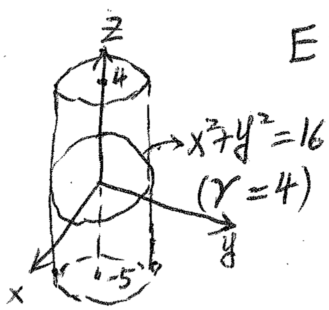
$$= \int_0^{\frac{\pi}{2}} \Big[-\tfrac{1}{2}\cos(x+2y)+\cos(x+y)\Big]_{y=0}^{y=x} dx$$

$$= \int_0^{\frac{\pi}{2}} \Big[-\tfrac{1}{2}\cos 3x+\cos 2x-\tfrac{1}{2}\cos x\Big] dx$$

$$= \Big[-\tfrac{1}{6}\sin 3x+\tfrac{1}{2}\sin 2x-\tfrac{1}{2}\sin x\Big]_{x=0}^{x=\pi/2}$$

$$= -\tfrac{1}{6}\sin\tfrac{3\pi}{2}+\tfrac{1}{2}\sin\pi-\tfrac{1}{2}\sin\tfrac{\pi}{2} = -\tfrac{1}{3}$$

Problem 2. (5 points) Using the cylindrical coordinates to compute the triple integral $\iiint_E \sqrt{x^2+y^2}\, dV$ where E is the solid that lies inside the cylinder $x^2+y^2=16$ and between the planes $z=-5$ and $z=4$.



$$E = \{(r,\theta,z) \mid 0\le\theta\le 2\pi,\ 0\le r\le 4,\ -5\le z\le 4\}.$$

$$\iiint_E \underbrace{\sqrt{x^2+y^2}}_{r}\,dV = \int_{-5}^{4}\int_0^{2\pi}\int_0^4 r\cdot r\,drd\theta dz$$

$$= \int_{-5}^{4}\int_0^{2\pi}\Big[\tfrac{r^3}{3}\Big]_{r=0}^{r=4} d\theta dz$$

$$= \int_{-5}^{4}\int_0^{2\pi}\tfrac{64}{3}\,d\theta dz$$

$$= \tfrac{64}{3}\cdot 2\pi\cdot(4-(-5))$$

$$= 384\pi$$